Vertical dilations

1 Consider the expression $\cos\theta$.

a Complete the table of values for different values of θ.

θ	0	60	90	120	180	240	270	300	360
$\cos\theta$									

b Sketch a graph of the function $y = \cos\theta$.

c State the maximum value of $\cos\theta$.

d State the minimum value of $\cos\theta$.

e State the range of values of $4\cos\theta$.

2 Consider the functions $y = \sin x$ and $y = \cos x$.

a State the domain of both functions.

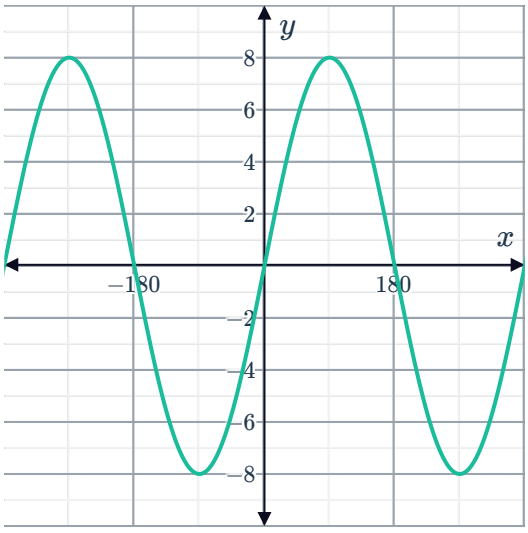
b State the range of both functions.

3 Consider the function $y = -2\cos x$.

a State the domain of the function.

b State the range of the function.

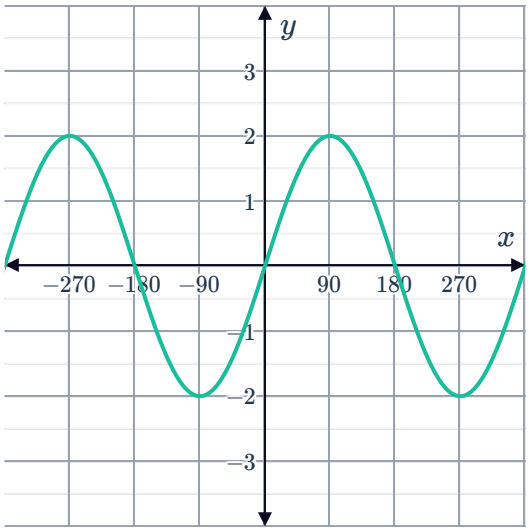
4 Consider the given graph of a function of the form $f(x) = A\sin x$:
State the amplitude of the function.



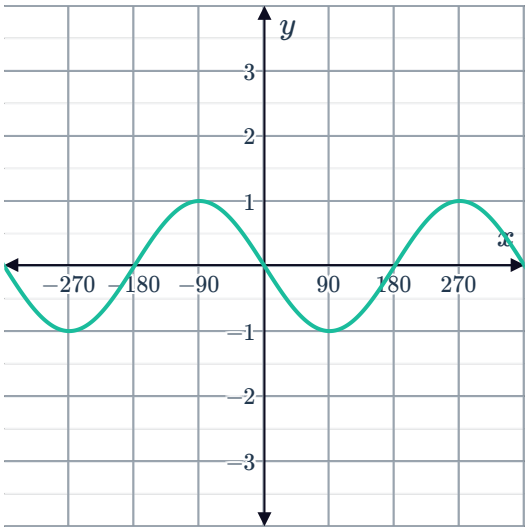
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5 State the equation of each of the functions graphed below given that they are of the form $y = a\sin x$ or $y = a\cos x$:

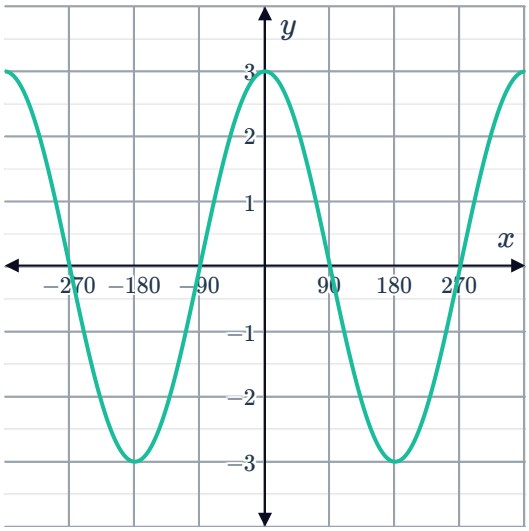
a



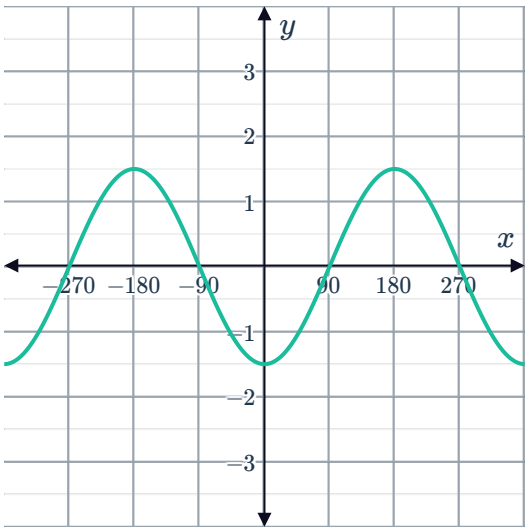
b



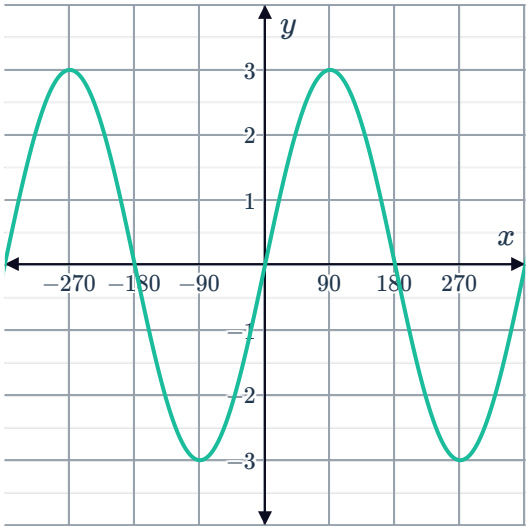
c



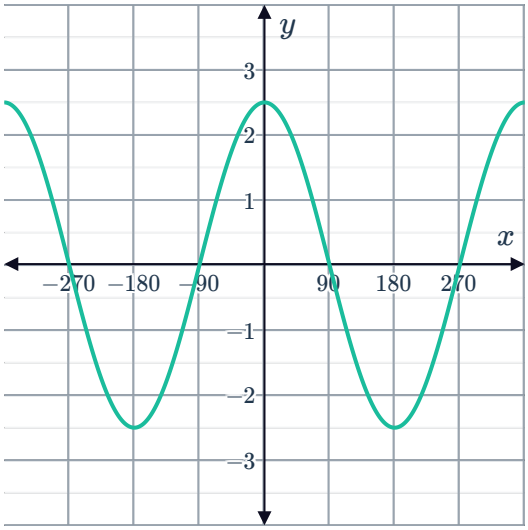
d



e

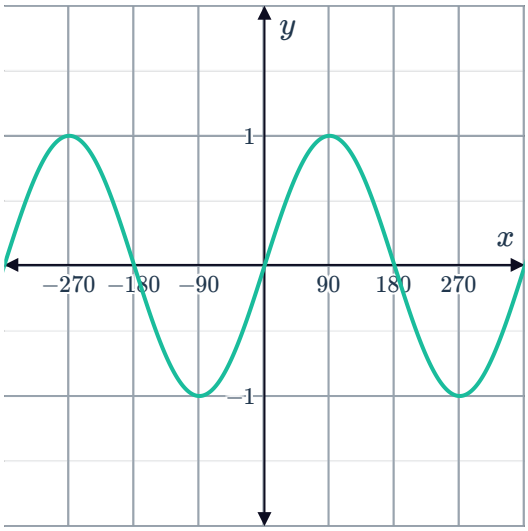


f



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6 Consider the graph of the function $y = \sin x$ for $0 \leq x < 360$:
At which value of x in the given domain would $y = -\sin x$ have a maximum value?



7 Consider the function $f(x) = -8\sin x$, where $0 \leq x \leq 180$.

a State the amplitude of the function.

b Find the value of $f(180)$.

c Find the minimum value of the function.

8 The function $y = k\sin x$ has a maximum value of 5. Find the value of k, where $k > 0$.

9 A sine function has the form $y = c\sin x$, a range of $[-2, 2]$ and a minimum at 90. Find an expression for y.

10 For each of following functions:

i State the amplitude.

ii Sketch a graph of the function.

a $y = 3\sin x$.

b $y = 4\cos x$

c $y = \frac{4}{3}\cos x$

d $y = -5\cos x$

e $y = \frac{5}{4}\sin x$

f $y = -4\sin x$

11 Consider the function $y = -3\cos x$.

a State the maximum value of the function.

b State the minimum value of the function.

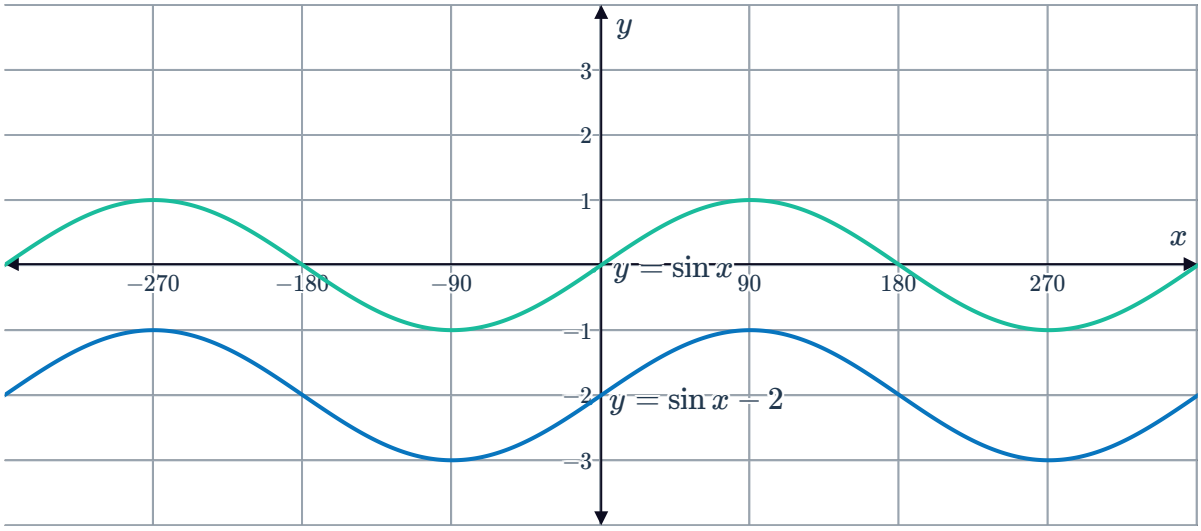
c State the amplitude of the function.

d Describe the two transformations required to obtain the graph of $y = -3\cos x$ from the graph of $y = \cos x$.

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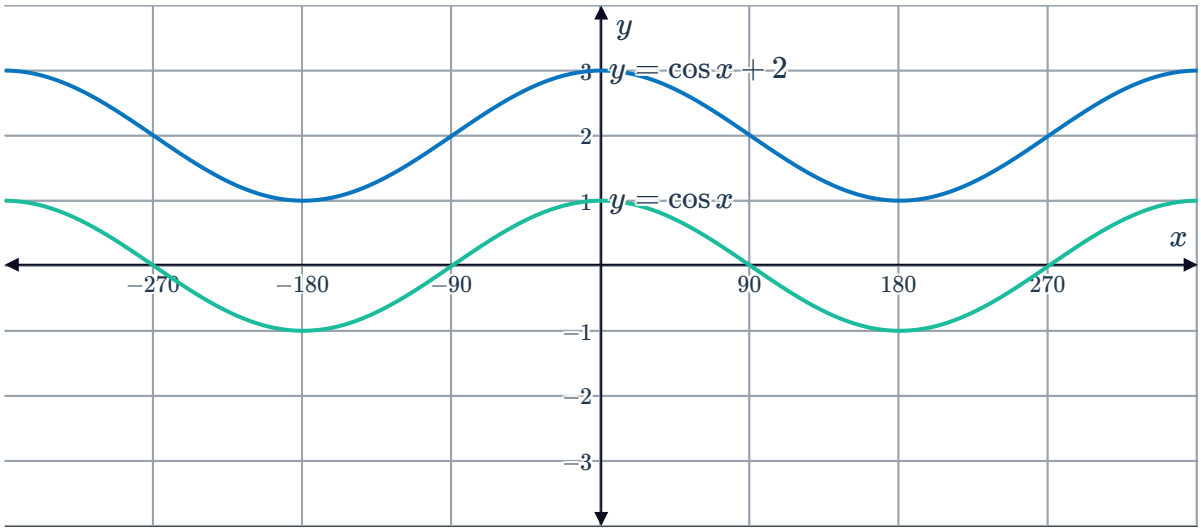
Vertical translations

12 Consider the two graphs $y = \sin x$ and $y = \sin x - 2$ below:



Describe the transformation required to obtain the graph of $y = \sin x - 2$ from $y = \sin x$.

13 Consider the two graphs $y = \cos x$ and $y = \cos x + 2$ below:



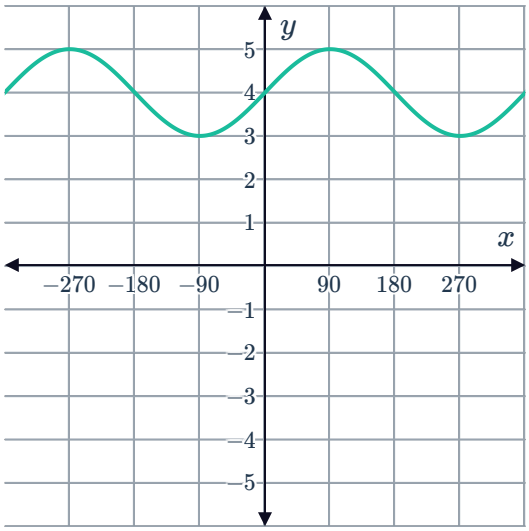
Describe the transformation required to obtain the graph of $y = \cos x + 2$ from $y = \cos x$.

14 Describe the transformation required to obtain the graph of $y = \sin x + 4$ from $y = \sin x$.

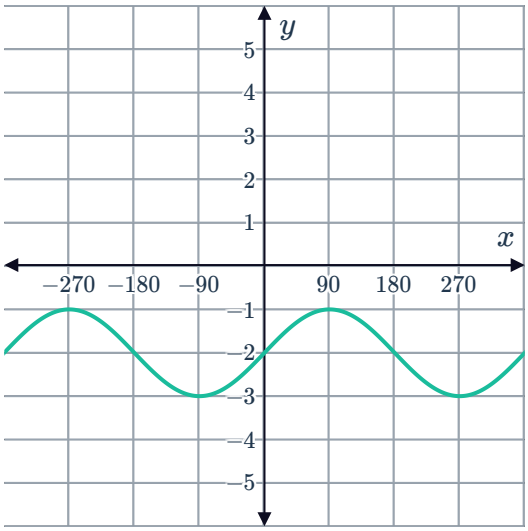
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15 State the equation of each of the functions graphed below given that they are of the form $y = \sin x + k$:

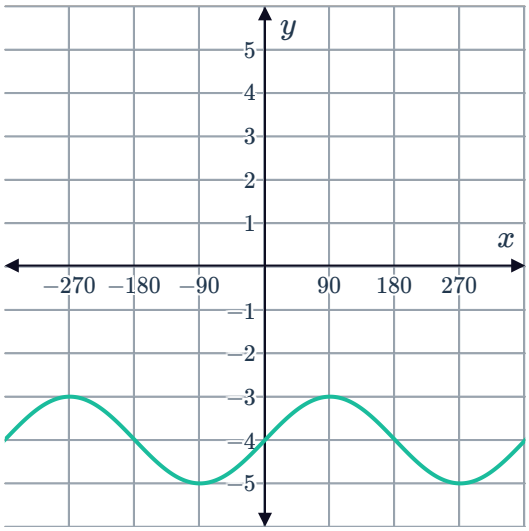
a



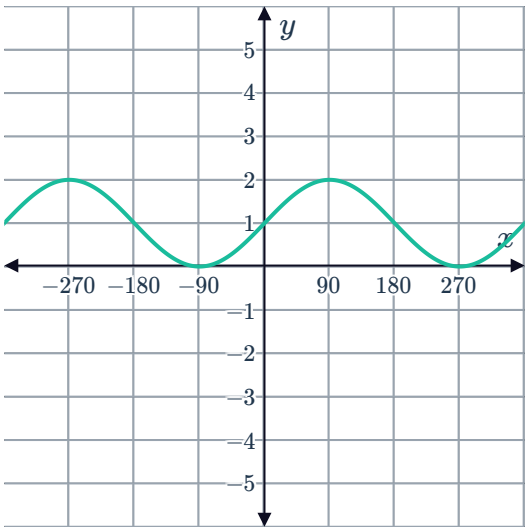
b



c



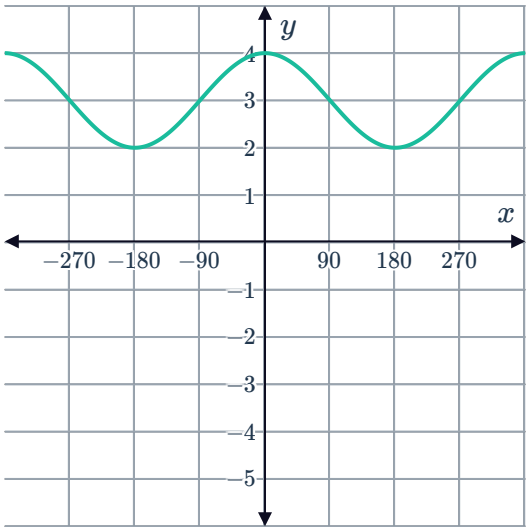
d



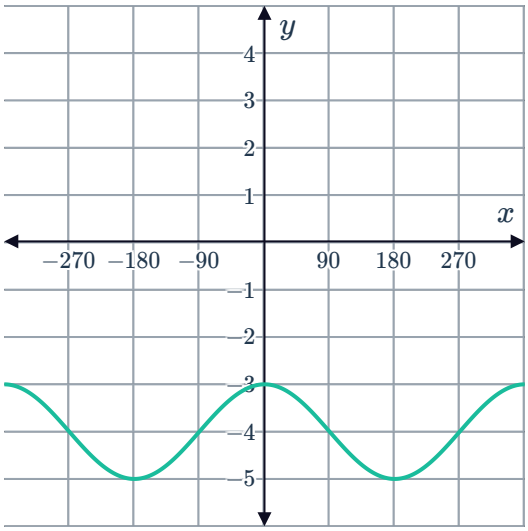
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16 State the equation of each of the functions graphed below given that they are of the form $y = \cos x + k$:

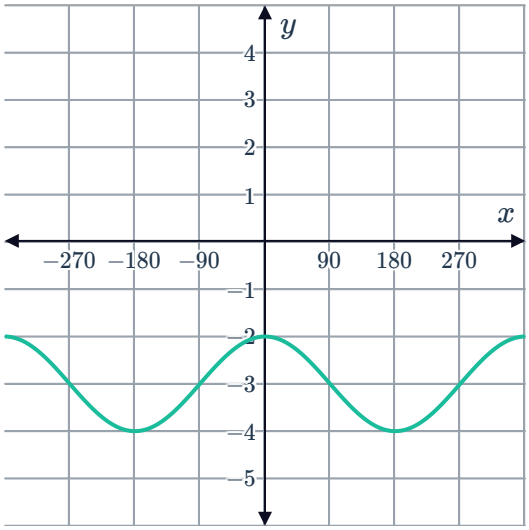
a



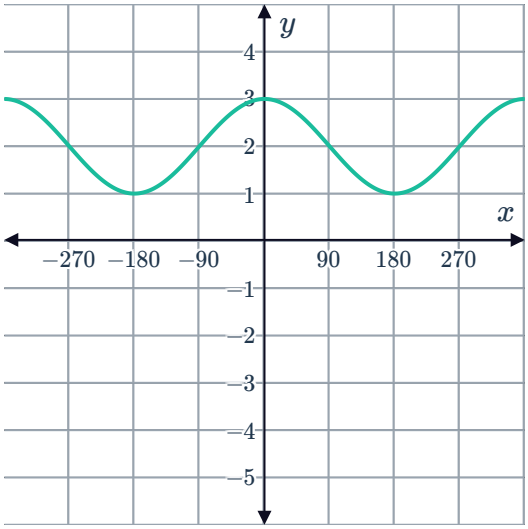
b



c



d



17 Describe the two transformations required to turn the graph of $y = \cos x$ into the graph of $y = -\cos x + 3$.

18 State the equation of the line that the function $y = \sin x + 2$ oscillates about.

19 Consider the function $y = \cos x + 4$.

a Describe the transformation required to obtain the graph of $y = \cos x + 4$ from $y = \cos x$.

b State the period of the function.

c Find the maximum value of the function.

d Find the minimum value of the function.

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20 For each of the functions below:

i State the period of the function.

ii State the amplitude of the function.

iii Find the maximum value of the function.

iv Find the minimum value of the function.

v Sketch a graph of the function for $-360 \leq x \leq 360$.

a $y = \sin x + 1$ **b** $y = \cos x - 3$ **c** $y = \sin x - 2$ **d** $y = 3\sin x + 2$

e $y = 3\cos x - 3$ **f** $y = 2\sin x - 3$

21 For each of the functions below:

i Find the value of y when $x = 90$.

ii State the amplitude of of the function.

iii Find the maximum value of the function.

iv Find the minimum value of the function.

v Describe the transformations required to obtain the graph of the function from $y = \cos x$.

vi Sketch a graph of the function for $0 \leq x \leq 360$.

a $y = 2\cos x + 3$ **b** $y = 2\cos x - 3$ **c** $y = -4\cos x + 2$ **d** $y = \frac{1}{3}\sin x + 2$

22 The function $y = \cos x + 5$ is translated 4 units up.

a State the equation of the new function after the translation.

b Find the maximum value of the new function.

23 For each of the functions below:

i Find the maximum value of the function. ii Find the minimum value of the function.

a $y = 2 - 3\sin x$ **b** $y = 4 - 3\sin x$ **c** $y = (\sin x)^2 + 1$ **d** $y = \frac{4}{3\cos x + 5}$

24 A sine function, y, has the form $y = c\sin x + d$ and a range of $[0, 4]$. Find an expression for y, where $c > 0$.

25 A cosine function, y, has the form $y = c\cos x - d$ and a range of $[-10, 6]$. Find an expression for y, where $c > 0$.

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